

MCU (ESP32-S3): Working Document

Digital Audio Player • Rev 0

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1	Summary	1
2	Design Requirements	1
3	External Components	1
4	Open Problems	2
5	Revision Log	2
6	Sources	2

1. Summary

Field	Value
Manufacturer / MPN	Espressif Systems, ESP32-S3 (base chip)
Package	56-QFN, 7 × 7 mm, exposed pad
Memory	No in-package flash or PSRAM; external SPI flash required
Function	Main controller: file decode, I2S to the DAC, UI, charger control, USB mass storage
Clock	External 40 MHz passive crystal (compulsory)

2. Design Requirements

Requirement	Value
Supply	+3V3_DIG
Storage	microSD over 4-bit SDMMC
Audio	I2S slave to the ES9038Q2M (DAC is master)
Control	I2C to the DAC and the BQ24250 at 100 kHz
Display	4-wire SPI to the ER-TFT024IPS-3
USB	USB-C, mass storage (GPIO19 / GPIO20)
Constraints	Low noise · Small footprint

3. External Components

Ref	Part	Notes	Qty
U	Espressif ESP32-S3	56-QFN, 7 × 7 mm	1
Y	HCI 0132M4-40.000F12FTNNK	40 MHz passive quartz crystal, AT-cut fundamental, 3.2 × 2.5 mm ceramic; ESR < 40 Ω (30 to 54 MHz); drive level 100 μW typ, 300 μW max	1

Crystal load capacitors (C1 = C2)

$$C_L = \frac{C_1 C_2}{C_1 + C_2} + C_{\text{stray}} \Rightarrow C_1 = C_2 = 2(C_L - C_{\text{stray}})$$

Source: standard crystal load equation; C_{stray} per the ESP32-S3 Hardware Design Guidelines.

4. Open Problems

#	Problem
1	Crystal: confirm load capacitance from the part-number suffix and size C1 / C2; follow the guideline reference circuit and layout.
2	External SPI flash selection, powered from VDD_SPI.
3	Memory: base chip without PSRAM, or add an external PSRAM.
4	LCD signals currently on flash pins (SPICS0, SPICLK, SPIQ); move to free GPIOs.
5	VDD_SPI tied to +3V3_DIG; connect per the reference design.
6	VDDA1 / VDDA2 on +3V3_ANALOG; move to +3V3_DIG.
7	I2C SDA and SCL crossed between the DAC and charger nets.
8	CHIP_PU RC delay capacitor missing.
9	Strapping pins: EN1_BQ (pull-up) on GPIO46 and CE_BQ on GPIO45; boot access on GPIO0.
10	Supply decoupling per pin group, per the guidelines.

5. Revision Log

Rev	Date	Change
0	2026-09-29	Initial issue: ESP32-S3 base chip, 40 MHz crystal, design references, schematic review items.

6. Sources

1. [ESP32-S3 Hardware Design Guidelines \(Espressif\)](#)
2. [ESP32-S3 Series Datasheet \(Espressif\)](#)
3. [ESP32-S3 Product Page \(DigiKey\)](#)
4. [0132M4 Series Crystal Datasheet \(HCL\)](#)